

ARC GENERALIZATION LAB

Concept Summary | Test-Time Adaptation & Benchmark Generalization

1. Problem & Central Claim

Public AI benchmarks such as ARC-AGI-1 have circulated online since 2019 and may appear in web-scraped LLM pretraining corpora. Therefore, performance on public tasks can conflate genuine few-shot rule induction with benchmark familiarity.

Central Claim: High performance on a public benchmark does not by itself establish generalization to novel tasks. A public-vs-fresh, distribution-matched comparison is a diagnostic test, **not proof of memorization**.

A performance drop on fresh tasks is a diagnostic signal consistent with five competing explanations: **benchmark familiarity, distribution mismatch, generator artifacts, task ambiguity, or baseline solver limits.**

2. Intervention: Distribution-Matched Fresh Generation

Using `pathwaycom/arc-task-gen`, the system synthesizes fresh ARC-style tasks anchored to public-task properties (grid size, color count, and pair count). **Joint constraint sampling** preserves natural feature covariance. Generated tasks undergo structural validation (cell values 0–9), semantic deduplication (cosine < 0.80), and public-evaluation similarity filtering (cosine < 0.92), producing an unseen, structurally matched evaluation split.

3. Test-Time Adaptation: Optimization vs. Context (BDH-CQ)

ARC provides demonstration pairs at inference time. We contrast two ways systems acquire the underlying rule:

Dimension	Optimization Route (HRM / TRM / TTT)	Contextual Recurrent Route (BDH-CQ)
Weights at test time	Modified ($\Delta W \neq 0$)	Fixed (W_{fixed}); $\nabla_W L = 0$
State evolution	Gradient descent on demonstration loss	Recurrent associative matrix accumulation: S_{t+1}
Compute profile	High latency; costly backward pass	Fast forward pass; reported \$0.0007/task
Primary risk	Overfitting sparse demonstrations	Finite associative state capacity

BDH-CQ (Dragon Hatchling, 150M parameters) adapts through an evolving recurrent associative-memory state during forward inference, with **zero test-time weight updates**. Reported public ARC-AGI-1 performance: **29.5% pass@2 at \$0.0007/task**.

Interactive sandbox: a 4×4 associative-state demo lets learners manipulate demonstration patterns, sample count T , and retention rate α using $S_{t+1} = \alpha S_t + (1-\alpha)\phi(x_t)\psi(y_t)^T$.

4. Evidence Discipline & Limitations

Evidence labels: **PUBLISHED RESULT** — **PATHWAY** (reported baselines); **MEASURED** (local execution); **DEMO DATA** (tutorial grids); **EDUCATIONAL TOY** (didactic 4×4 matrix).

Limitations: Generated-task solvability is not programmatically guaranteed (addressed through a blind human-audit module). Distribution matching captures measurable grid statistics, not the full set of human cognitive priors.

5. Primary Literature

- Engdahl et al. (2026) — *BDH-CQ: In-Context Learning with Recurrent Latent Reasoning*, arXiv:2608.09888.
- Bordes et al. (2024) — *Gemini’s ARC-AGI Capabilities and Contamination*, arXiv:2407.00645.
- Akyürek et al. (2024) — *Effectiveness of Test-Time Training for Abstract Reasoning*, arXiv:2411.07279.
- Chollet (2019) — *On the Measure of Intelligence*, arXiv:1911.01547.

Public Demonstration: ARC Generalization Lab — famous-sable-446bd6.netlify.app